WORKED SOLUTION

STEP III 2013, Question 9

Mechanics • SHM

STEP III 2013, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q9

The key idea

The submerged part of the sphere is a spherical cap, and its volume comes from slicing it into discs, which is a volume of revolution. After that the question is Newton's second law with buoyancy as the only force besides weight. For small oscillations we put $x = x_0 + y$, linearise the cubic on the right about the equilibrium, and read off ω^2 from an equation of the form $\ddot{y} = -\omega^2 y$.

Solution

The displaced volume

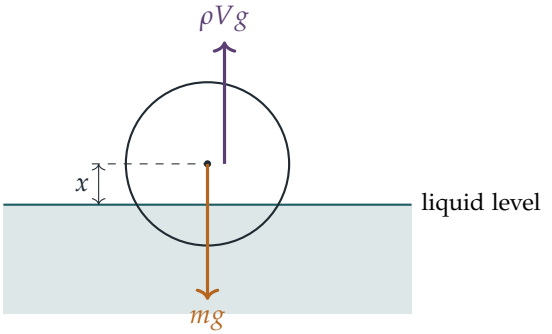
Take the centre of the sphere as origin with z measured upwards. The liquid surface is then the plane $z = -x$, and the submerged part is the part of the ball with $-R \le z \le -x$. This makes sense for $-R < x < R$, so it covers the case in the question. At height z the cross-section is a disc of radius $\sqrt{R^2 - z^2}$, so

$$\begin{aligned} V &= \int_{-R}^{-x} \pi\left(R^2 - z^2\right) \mathrm{d}z = \pi\left[R^2 z - \tfrac{1}{3}z^3\right]_{-R}^{-x} \\ &= \pi\left(-R^2 x + \tfrac{1}{3}x^3 + R^3 - \tfrac{1}{3}R^3\right) = \frac{\pi}{3}\left(2R^3 - 3R^2 x + x^3\right), \end{aligned}$$

which is the required volume.

The equation of motion

Model the sphere as a rigid body moving vertically. Because the tank is large, the liquid level does not move as the sphere bobs, so $\ddot{x}$ is the upward acceleration of the sphere. Its mass is $m = \frac{4}{3}\pi R^3 \rho_s$. Two forces act on it: its weight mg downwards, and the buoyancy force $\rho V g$ upwards.



Resolving vertically, with upwards positive,

$$m\ddot{x} = \rho V g - mg \quad \Longrightarrow \quad m\,(g + \ddot{x}) = \rho g V.$$

Substituting for m and V:

$$\tfrac{4}{3}\pi R^3 \rho_s\,(g + \ddot{x}) = \tfrac{\pi}{3}\left(2R^3 - 3R^2 x + x^3\right)\rho g,$$

and multiplying by $3/\pi$ gives

$$4R^3 \rho_s\,(g + \ddot{x}) = \left(2R^3 - 3R^2 x + x^3\right)\rho g,$$

as required.

The equilibrium density

In equilibrium $\ddot{x} = 0$. Putting $x = \frac{1}{2}R$:

$$4R^3 \rho_s g = \left(2R^3 - \tfrac{3}{2}R^3 + \tfrac{1}{8}R^3\right)\rho g = \tfrac{5}{8}R^3 \rho g,$$

so

$$\rho_s = \frac{5}{32}\,\rho.$$

This is less than $\frac{1}{2}\rho$, as it should be: the centre is above the surface, so less than half the sphere is under water.

Small oscillations

Write $f(x) = 2R^3 - 3R^2 x + x^3$, and put $x = \frac{1}{2}R + y$ with y small. Then $\ddot{x} = \ddot{y}$.

> **Ask yourself** How do we turn the cubic into something linear in y?

We expand f about the equilibrium and keep only first-order terms: $f(\frac{1}{2}R + y) = f(\frac{1}{2}R) + f'(\frac{1}{2}R)\,y + O(y^2)$. Here

$$f'(x) = -3R^2 + 3x^2, \qquad f'(\tfrac{1}{2}R) = -3R^2 + \tfrac{3}{4}R^2 = -\tfrac{9}{4}R^2,$$

and $f(\frac{1}{2}R) = \frac{5}{8}R^3$ from before. Using $\rho_s = \frac{5}{32}\rho$, the left-hand side of the equation of motion is $4R^3 \cdot \frac{5}{32}\rho\,(g + \ddot{y}) = \frac{5}{8}R^3\rho\,(g + \ddot{y})$. So, to first order,

$$\tfrac{5}{8}R^3\rho\,(g + \ddot{y}) = \left(\tfrac{5}{8}R^3 - \tfrac{9}{4}R^2 y\right)\rho g.$$

The $\frac{5}{8}R^3\rho g$ terms cancel (that is just the equilibrium condition), leaving

$$\tfrac{5}{8}R^3\ddot{y} = -\tfrac{9}{4}R^2 g\, y \quad \Longrightarrow \quad \ddot{y} = -\frac{18g}{5R}\, y.$$

This is simple harmonic motion with $\omega^2 = \dfrac{18g}{5R}$, so the period of small oscillations is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{5R}{18g}}.$$

Checks

- Volume: at $x = R$ the formula gives 0 (just touching the surface), at $x = 0$ it gives $\frac{2}{3}\pi R^3$ (half the sphere), and at $x = -R$ it gives $\frac{4}{3}\pi R^3$ (fully submerged).
- Dimensions: $\omega^2 = 18g/(5R)$ has units s^{-2}, as it should.
- Numbers: with $R = 1\,\mathrm{m}$ and $g = 9.81\,\mathrm{m\,s^{-2}}$ the period is about 1.06 s. Stepping the full nonlinear equation forward by computer from a small displacement gives the same period to better than one part in 10^4.

Common slips

- Taking the height of the submerged cap as x rather than $R - x$: x is the height of the centre *above* the liquid.
- Forgetting that the restoring force comes from $f'(\frac{1}{2}R)$, which is negative. A sign error here gives $\ddot{y} = +\omega^2 y$, which isn't oscillation at all.
- Dropping the sphere's own mass on the left: $\ddot{x}$ multiplies ρ_s, not ρ. (Real floating bodies also drag some liquid with them, the "added mass", which the question tells us to ignore.)

Beyond STEP

The method used here, expanding the force about an equilibrium and keeping the linear term, is the standard way to find small oscillations of any system. Its sign gives stability, and its size gives the frequency. In a first-year course this becomes linearisation of $\ddot{x} = F(x)$, and then the study of normal modes when there are several degrees of freedom. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers small oscillations about equilibrium and fluid buoyancy at this level.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.